

National Institute of Technology, Rourkela

Department of Computer Science and Engineering

B. Tech/M. Tech.(IS)/Ph.D./PhD(Ex.) End-Semester Examination (Autumn) 2022

Subject: **Foundations of Computer Security**

Subject Code: **CS 6301**

Full Marks: **50**

Duration: **3 Hours**

Answer **ALL** questions.

Figures at the right margin indicate marks.

Question paper contains TWO pages. All parts of a question must be answered at one place.

Use of non-programmable calculator is allowed

1. (a) Prof. Wang purchased systems for his laboratory. He desires to make a laboratory by grouping them. If he makes them into 5 equal groups, 4 are left over. If he makes them into 8 equal groups, 6 are left over. If he makes them into 9 equal groups, 8 are left over. What is the smallest number of systems that Prof. Wang could have? [4]
(b) Callie wants to use RSA encryption scheme to send the message 13 to Alice using Alice's public keys. Alice's public keys are (33,3). Calculate the cipher text, and also show how Alice recovers the message. [3]
(c) Prove that for any prime p , we have $(x + y)^p \equiv x^p + y^p \pmod{p}$. [3]
2. (a) Define bit, byte, word, block and state. With suitable block diagram, explain the various transformation that takes place in each round of AES. [4]
(b) Find the result of the following using square-and-multiply method. [3]
$$18^{13} \bmod 107$$

(c) Write the Miller Rabin primality test algorithm. Check whether the integer 271 pass the test. [3]
3. Consider a block cipher, called A, shown in the table below. The table gives the ciphertext C produced when encrypting the plaintext P with one of the four keys. [10]

Table 1: Cipher A

Plain text		Key		
P	00	01	10	11
0000	1111	0000	0101	0001
0001	0001	0010	1001	0111
0010	1010	0101	0111	1000
0011	0111	1010	0010	1111
0100	1000	1001	1100	0101
0101	1100	1110	1011	1010
0110	1011	0111	1110	0100
0111	0000	1111	0001	1110
1000	1110	0001	1101	0110
1001	1001	0011	1000	1011
1010	0100	1100	0000	1101
1011	0110	1101	0100	1001
1100	0101	0100	0110	0010
1101	1101	0110	1111	0000
1110	0010	1000	0011	1100
1111	0011	1011	1010	0011

Using cipher A and one of the following modes of operation, encrypt the plain text P with key K:

P 1101 0100 1100 0100

K 01

In all cases assume any initial values are 0. What will be the output if the following mode of operations are used?

- (a) Counter (increment by one)
 - (b) CBC
4. (a) Let Peter and Ali both decided to use Rabin Crypto-system to exchange their messages with n as 2021 and p as 43. Peter receives a cipher text 1821 from Ali. How Peter will decrypt it and guess the actual message. (*Hint: message only contains English alphabet [a-z] with a mapping to [0-25]*). [4]
- (b) Given a super-increasing sequence 7, 11, 23, 43, 87, 173, 357 with 1001 as modulus and the value of $r = 31$. The permutation table is [7,5,2,4,3,1,6]. Encrypt and decrypt the message 0101110. [4]
- (c) Is it possible to get the value of p and q for a given $n(= p \times q)$ and $\Phi(n)$? If yes then how can we find the factors p and q . [2]
5. (a) What is Elliptic Curve? Generate all points for the curve defined as $E_{13}(3, 2)$. If $(2, 9)$ is a point on the curve then find $3 * (2, 9)$. [4]
- (b) Factor 31861 using the Pollard Rho method with polynomial $f(x) = x^2 + 1$ and initial guess $x_0 = 1$. [4]
- (c) Define Euler's Totient function. [2]

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